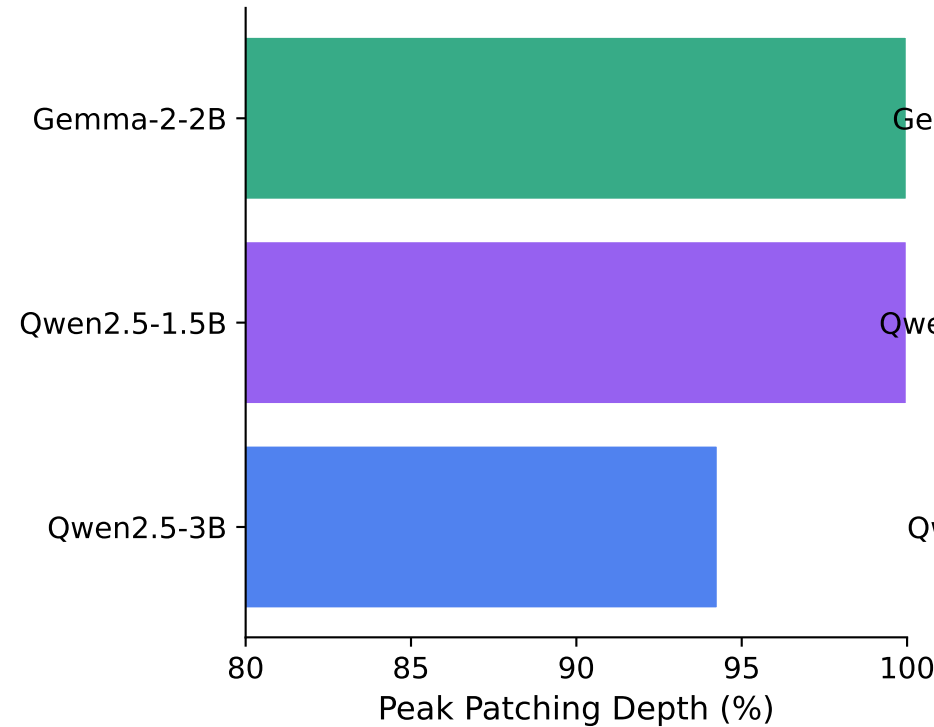
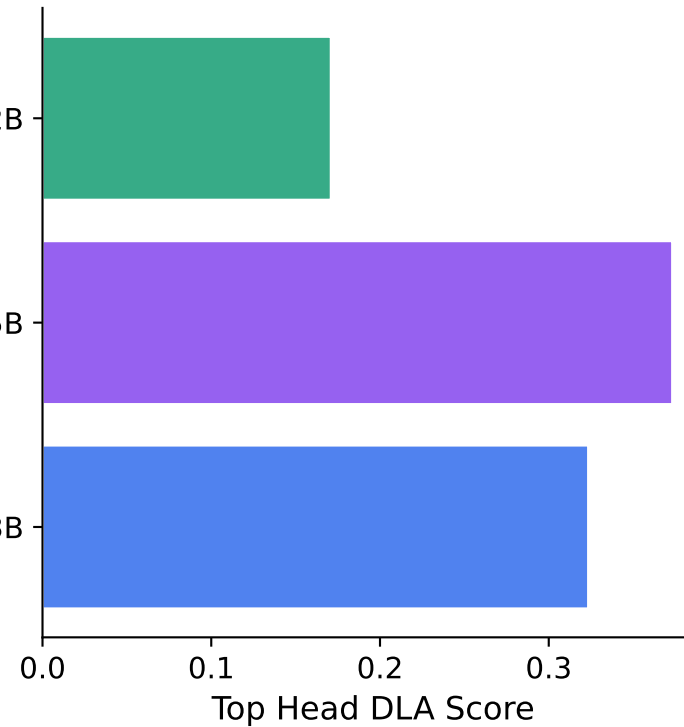


Source Arbitration: Cross-Model Mechanistic Analysis

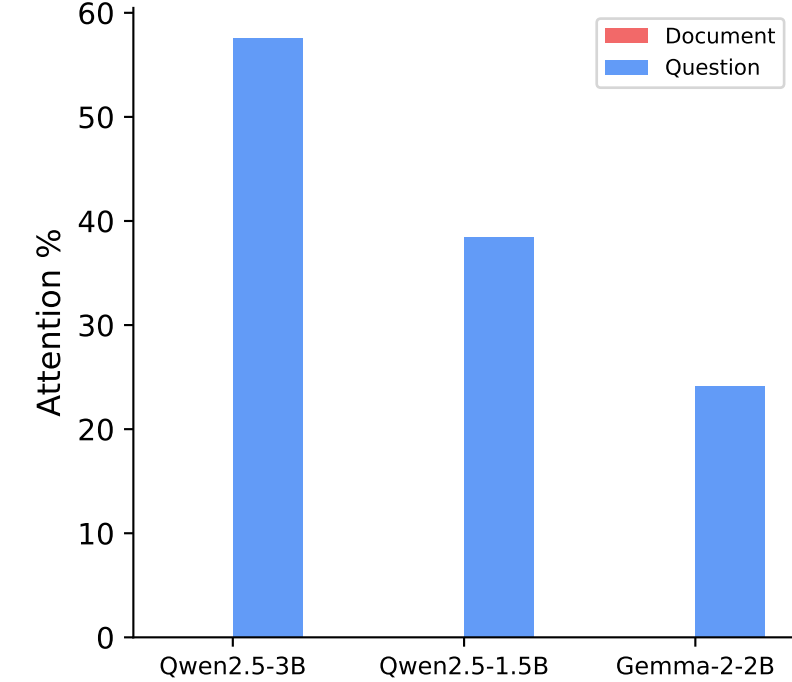
A. Circuit Location



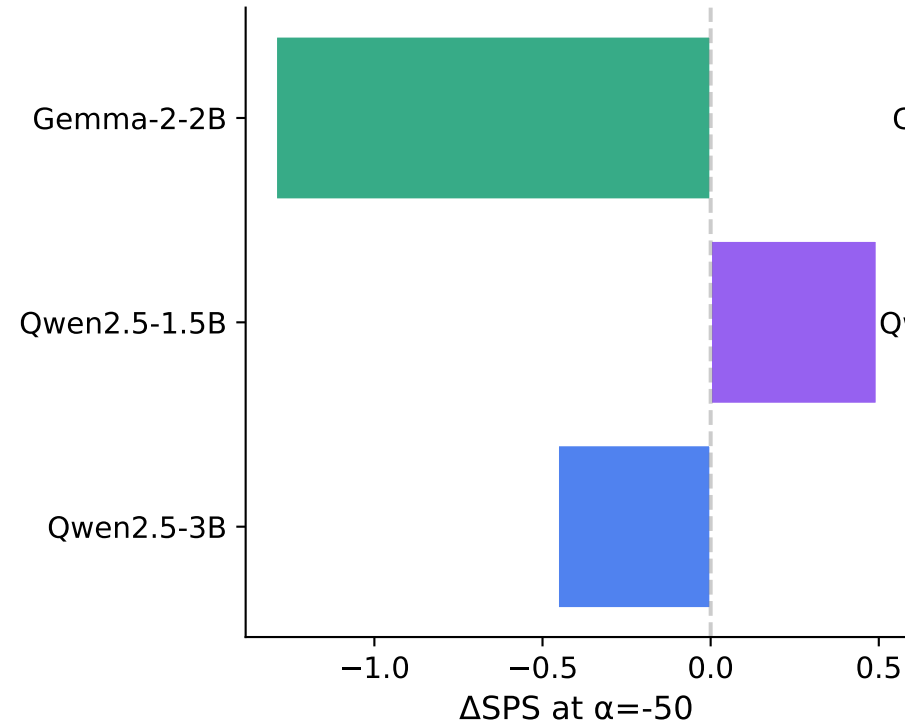
B. Max Logit Attribution



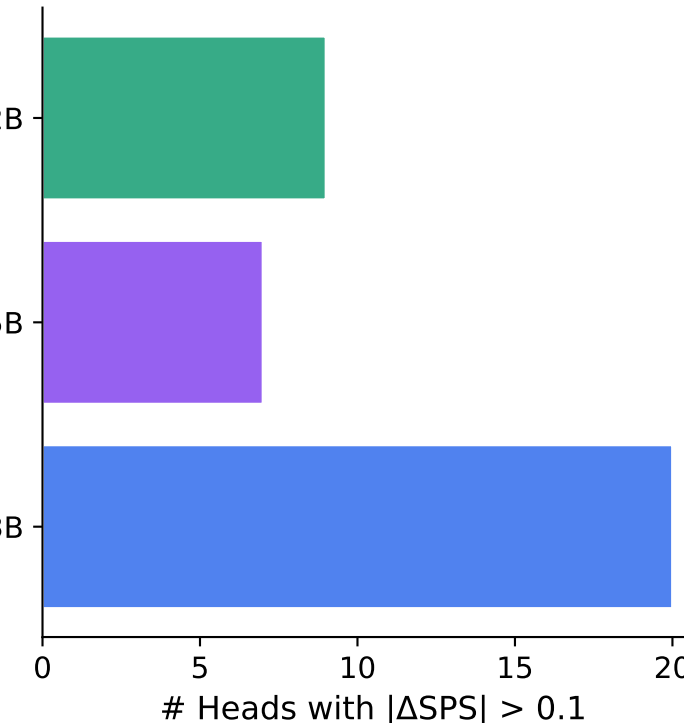
C. What Heads Attend To



D. Steering Effect



E. Circuit Sparsity



F. Summary

Key Findings:

- Context-override circuit at 92-96% depth (last 2-3 layers)
- Heads read question (50-74%), NOT document (0%)
- Qwen-1.5B has memory-defense heads (inverted DLA & steering)
- Gemma uses distributed circuit (many weak heads vs few strong)